Worksheet 7A: Introduction to Matrices

Series 7: Linear Algebra
AIML Foundations Mathematics
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What This Worksheet Is About
A matrix is a rectangular grid of numbers — and it turns out to be one of the most powerful ideas in mathematics. Matrices let us represent entire systems of equations as a single object, describe geometric transformations, store datasets, and run the computations inside every neural network. In this worksheet you will learn to read, write, add, multiply, and invert matrices.

Part A: Reading and Writing Matrices (8 problems)

A matrix with m rows and n columns is called an $m \times n$ **matrix**. The element in row i, column j is written a_{ij}.

$$A = \begin{bmatrix} 2 & -1 & 5 \\ 0 & 3 & -4 \end{bmatrix} \quad \text{(a } 2 \times 3 \text{ matrix)}$$

Let $A = \begin{bmatrix} 4 & 7 & -2 \\ 1 & 0 & 6 \\ -3 & 5 & 8 \end{bmatrix}$

1. State the dimensions of A.
2. Find: (a) a_{11} (b) a_{23} (c) a_{32} (d) a_{33}
3. Write down the second **row** of A as a row vector.
4. Write down the third **column** of A as a column vector.
5. A dataset has 4 training examples and 3 features. Write a 4×3 feature matrix X with made-up values.
6. Write out the identity matrix I_3 (a 3×3 matrix with 1s on the diagonal and 0s elsewhere).
7. Write out a 2×3 zero matrix O.
8. (a) Which of the following are square? 2×2, 3×2, 4×4, 1×3. (b) Write an example 3×3 diagonal matrix.

Part B: Addition, Subtraction, and Scalar Multiplication (12 problems)

Addition requires identical dimensions. Scalar multiplication multiplies every entry by k.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}, \qquad k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

Let $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ -3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$

9. Compute $A + B$.
10. Compute $A - C$.
11. Compute $3A$.
12. Compute $2B - A$.
13. Compute $A + B + C$.
14. Is $A + B = B + A$? Verify.
15. The **transpose** A^T swaps rows and columns. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, find A^T and state its dimensions.
16. Verify that $S = \begin{bmatrix} 1 & 4 & 7 \\ 4 & 2 & 5 \\ 7 & 5 & 3 \end{bmatrix}$ is symmetric (i.e. $S^T = S$).

AI Connection: Gradient Updates

17. In training, weights are updated by $W_{\text{new}} = W_{\text{old}} - \alpha G$ where $\alpha = 0.1$. Given $W_{\text{old}} = \begin{bmatrix} 0.5 & -0.3 \\ 1.2 & 0.8 \end{bmatrix}$, $G = \begin{bmatrix} 0.4 & -0.2 \\ 0.6 & 1.0 \end{bmatrix}$, compute W_{new}.
18. Why does the formula *subtract* αG rather than add it?
19. If A is 3×4 and B is 3×4, what are the dimensions of $A + B$ and $2A - 3B$?
20. If A is 2×3 and B is 3×2, can you compute $A + B$? Explain.

Part C: Matrix Multiplication (14 problems)

To multiply A ($m \times n$) by B ($n \times p$): the inner dimensions must match. The result is $m \times p$. Element c_{ij} is the dot product of row i of A with column j of B:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Example: $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 3 \\ 6 & 15 \end{bmatrix}$

21. Before multiplying, check dimensions. State whether each product is defined and the result's size: (i) 2×3 by 3×4 (ii) 3×2 by 3×2 (iii) 4×1 by 1×3 (iv) 1×4 by 4×1
22. Compute AB where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 0 \\ 1 & -1 \end{bmatrix}$.

23. Compute BA with the same matrices. Is $AB = BA$?

24. Compute AI where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. What do you notice?

25. Let $A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 0 & -2 \\ 3 & 1 \end{bmatrix}$. Compute AB.

26. Let $\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$. Compute $A\mathbf{x}$.

AI Connection: The Forward Pass

27. A neural network layer computes $\mathbf{y} = W\mathbf{x} + \mathbf{b}$. Given $W = \begin{bmatrix} 0.2 & 0.8 \\ -0.5 & 0.3 \\ 0.1 & 0.6 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0.1 \\ -0.2 \\ 0.3 \end{bmatrix}$. Compute $W\mathbf{x}$ and then $W\mathbf{x} + \mathbf{b}$.

28. What are the dimensions of W, $\mathbf{x}$, and $\mathbf{y}$? If the next layer has 2 output neurons, what would the next weight matrix dimensions be?

29. Write each system as a matrix equation $A\mathbf{x} = \mathbf{b}$: (a) $3x - 2y = 5$, $x + 4y = -3$ (b) $x + y + z = 6$, $2x - y + 3z = 11$, $x + 2y - z = 2$

30. Verify $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ solves system (a) by computing $A\mathbf{x}$.

31. Compute $(A + B)C$ and $AC + BC$ for $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$. What property does this show?

32. True or false: (a) $AB = BA$ always. (b) $(AB)C = A(BC)$ always. (c) $AI = A$ always.

33. Feature matrix X is 100×5, weight vector $\mathbf{w}$ is 5×1. What is the shape of $\hat{\mathbf{y}} = X\mathbf{w}$?

34. Apply rotation matrix $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Does it rotate 90° anticlockwise?

Part D: Determinants and Inverses (10 problems)

For a 2×2 matrix: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.

Inverse: $A^{-1} = \dfrac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (only when $\det A \neq 0$).

For a 3×3 matrix, expand along the first row (cofactor expansion).

35. Compute the determinant of each: (a) $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & -1 \\ 10 & -2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 4 & -2 & 1 \end{bmatrix}$

36. For (a) in problem 35, find A^{-1}.

37. Verify $AA^{-1} = I$.

38. Matrix (b) in problem 35 has det = 0. Explain geometrically why no inverse exists.

39. Solve $A\mathbf{x} = \mathbf{b}$ via $\mathbf{x} = A^{-1}\mathbf{b}$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$.

40. What does "ill-conditioned" mean, and why is it a problem when training ML models?

41. A matrix scales x by 3 and y by 2. Write it, find its determinant, and explain what det says about area scaling.

42. If $\det A = 5$ and $\det B = 3$ (both 2×2), what is $\det(AB)$?

43. Find the trace of: (a) $\begin{bmatrix} 4 & 2 \\ -1 & 7 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 5 & -1 \\ 0 & 4 & 2 \end{bmatrix}$

44. If $\mathrm{tr}(A) = 5$ and $\det A = 6$, find the eigenvalues λ_1, λ_2 (using $\lambda_1 + \lambda_2 = 5$, $\lambda_1\lambda_2 = 6$).

Part E: Matrices in Context (6 problems)

45. Confusion matrix $C = \begin{bmatrix} 850 & 30 & 20 \\ 15 & 920 & 25 \\ 10 & 20 & 970 \end{bmatrix}$ (rows = true; columns = predicted; Cat, Dog, Bird). (a) Cats correctly classified? (b) Dogs misclassified as birds? (c) Overall accuracy?

46. Covariance matrix $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. (a) Standard deviations? (b) Why must a covariance matrix be symmetric?

47. Apply softmax to $\mathbf{z} = \begin{bmatrix} 2.0 \\ 1.0 \\ 0.5 \end{bmatrix}$. Use $e^2 \approx 7.39$, $e^1 \approx 2.72$, $e^{0.5} \approx 1.65$.

48. Which are valid stochastic vectors? (a) $[0.3, 0.5, 0.2]^T$ (b) $[0.6, -0.1, 0.5]^T$ (c) $[1, 0, 0]^T$ (d) $[0.4, 0.4, 0.3]^T$

49. Give an example 2×2 doubly stochastic matrix.

50. Reflection: (a) Where does this worksheet connect to simultaneous equations (ws03b)? (b) Where does the dot product from ws02a appear? (c) Why does gradient descent require matrix operations at scale?